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MILITARY SPECIFICATION

POWER SUPPLY, RECTIFIER
(For Torpedo Mark 37 Mods 2 and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement, inspection, and test of a rectifier power supply, NAVORD LD 162564.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods for
MIL-C-24308	Connectors, Electric (Part Number M24308/3-2) and M24308/1-2)
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

FSC-1355

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DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 162564	Rectifier Power Supply Assembled, and All Parts and Subassemblies Listed Thereon
NAVORD LD 495864	Shipping (Packing) Condition for One Rectifier Power Supply Assembled
NAVORD Dwg 738826	Schematic Wiring Diagram

PUBLICATIONS

Naval Ordnance Systems Command

NAVORD OS 6341	Description and Requirements for Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The rectifier power supply is a device which produces various low power level dc voltages for

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use by the torpedo dynamic control circuits. Rectifier power supply input power is taken from the torpedo 115-volt root-mean-square (rms), three phase, 400 Hertz (Hz) ac supply. The rectifier power supply consists of a three-phase transformer, the secondary windings of which feed separate silicon rectifier banks and their associated filter circuits. The output voltages are furnished in two separate ungrounded sections. One section provides plus 300 volts dc and plus 150 volts dc and the other section provides plus 150 volts dc and minus 90 volts dc.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the rectifier power supply shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing rectifier power supplies that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the rectifier power supply shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 162564.

3.2.4 Finishes, coatings, construction, and dimensions. The finishes, coatings, construction, and dimensions for the rectifier power supply shall be strictly in accordance with applicable drawings, specifications and publications listed in LD 162564.

3.2.5 Moisture, fungus, and corrosion. The rectifier power supply shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements on the drawings and specifications listed in OS 6341.

3.2.6 Marking. The marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 162564.

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3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or the appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Performance and product characteristics. The rectifier power supply shall have the product characteristics and meet the performance requirements given below:

3.3.1 Input

3.3.1.1 Voltage. Three phase, 115 volts plus or minus 1 volt rms line-to-line average with the difference between any two line-to-line voltages not exceeding 2.5 volts rms.

3.3.1.2 Frequency. 400 Hz plus or minus 2 percent.

3.3.1.3 Current. 0.42 ampere rms per line maximum.

3.3.2 Output. The rectifier power supply shall have a total of four output circuits. The output circuits shall be contained within two mutually isolated output sections.

3.3.2.1 Section 1. Section 1 shall consist of two dc circuits, the characteristics of which are specified below. The negative terminal shall be common to both circuits.

3.3.2.1.1 300 volt dc circuit

3.3.2.1.1.1 Output load voltage. 300 plus or minus 15 volts dc.

3.3.2.1.1.2 Output current. 30 plus or minus 2 milliamperes (ma) dc.

3.3.2.1.1.3 Output impedance. 20 ohms maximum at 400 Hz.

3.3.2.1.1.4 AC voltage content. The ac voltage content in the output load voltage in a frequency band from 390 Hz to 150 kiloHertz (kHz) per second shall not exceed 3×10^{-3} volts rms.

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3.3.2.1.2 150 volt dc circuit.

3.3.2.1.2.1 Output voltage. 150 plus or minus 7.5 volts dc.

3.3.2.1.2.2 Output current. 80 plus or minus 4 ma, dc.

3.3.2.1.2.3 Output impedance. 100 ohms maximum at 400 Hz.

3.3.2.1.2.4 AC voltage content. The ac voltage content in the output load voltage in a frequency band from 390 Hz to 150 kHz shall not exceed 50 x 10^{-3} volts rms.

3.3.2.1.2.5 Arc suppressor. An arc suppressor shall be connected between the positive 150 volt dc line and a terminal on the rectifier power supply receptacle. Design of the arc suppressor and its connection within the rectifier power supply shall be as specified on Drawing 738826.

3.3.2.2 Section 2. Section 2 shall consist of two dc circuits, the characteristics of which are specified below. The negative terminal of 3.3.2.2.1 shall be common with the positive terminal of 3.3.2.2.2.

3.3.2.2.1 150-volt dc circuit

3.3.2.2.1.1 Open circuit output voltage. 165-volts dc maximum.

3.3.2.2.1.2 Output load voltage. 150 plus or minus 1 volt dc.

3.3.2.2.1.3 Output current. 125 plus or minus 5 ma, dc.

3.3.2.2.1.4 Output impedance. 9 ohms maximum at 60 kHz; 9 ohms maximum at 7 kHz; 9 ohms maximum at 1 kHz.

3.3.2.2.1.5 AC voltage content. The ac voltage content in the output load voltage with the output or load current adjusted to 90 plus or minus 5 ma shall not exceed the following:

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390 Hz to 150 kHz: 0.2 volt rms.
0.9 to 1.1 kHz: 2.0×10^{-3} volts rms.
6 to 8 kHz: 0.2×10^{-3} volts rms.

3.3.2.2.2 90-volt dc circuit

3.3.2.2.2.1 Open circuit output voltage. 105 volts dc maximum.

3.3.2.2.2.2 Output load voltage. 90 plus or minus 10 volts dc.

3.3.2.2.2.3 Output current. 1.5 plus or minus 0.1 ma dc.

3.3.2.2.2.4 Output impedance. 100 ohms maximum at 60 kHz.

3.3.2.2.2.5 AC voltage content. The ac voltage content in the open circuit voltage shall not exceed the following:

390 Hz to 150 kHz: 2.0 volts rms.
55 to 65 kHz: 2.0×10^{-3} volts rms.

3.3.2.2.2.6 Energy absorption. The 90-volt dc circuit shall be capable of absorbing 80 plus or minus 5 milliwatt-seconds of energy during 33 plus or minus 3 milliseconds (ms) periods. The absorption periods shall occur at a rate of one period every 1250 plus or minus 8 ms. The output load voltage shall not change more than 40 volts during any absorption period.

3.3.3 Insulation requirements

3.3.3.1 Dielectric requirements. The rectifier power supply shall be capable of withstanding 1000 volts rms at 60 Hz for one minute, applied between the following points without any arcing or breakdown of insulation. Prior to applying test voltages, each of the 14 rectifiers is to be shorted by means of jumper cables. Capacitors shall not be wired in during dielectric tests.

3.3.3.1.1 Transformer primary. Between all H terminals of the transformer simultaneously and the chassis with all pins of the male connector MIL-C-24308(3-2), except those connected to the transformer primary windings, grounded to the rectifier power supply chassis.

3.3.3.1.2 Section 1. Between all terminals of Section 1 and the chassis, with all pins of the male connector MIL-C-

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24308(3-2) except those connected to the circuits of Section 1 simultaneously grounded to the rectifier power supply chassis.

3.3.3.1.3 Section 2. Between all terminals of Section 2 and the chassis, with all pins of the male connector MIL-C-24308 (3-2) except those connected to the circuits of Section 2 simultaneously grounded to the rectifier power supply chassis.

3.3.3.2 Insulation resistance. The rectifier power supply shall have a minimum of 30 megohms resistance at applicable dc between the following points:

3.3.3.2.1 Section 1. Between all terminals of Section 1 simultaneously and the chassis, with all terminals of Section 2 grounded to the rectifier power supply chassis. Test voltage is 400 vdc.

3.3.3.2.2 Section 2. Between all terminals of Section 2 simultaneously and the chassis, with all terminals of Section 1 grounded to the rectifier power supply chassis. Test voltage is 200 vdc.

3.3.4 Operating time. The rectifier power supply shall be capable of operating under full load continuously for a period of 85 plus or minus 5 minutes, followed by sufficient time for the unit to cool to ambient temperature before operating again.

3.3.5 Operating life. The rectifier power supply shall have a useful operating life of at least 200 hours. During the life of the rectifier power supply, maintenance shall not be permitted.

3.3.6 Warm-up time. The output voltage and ac voltage content of the output voltage shall be within the limits specified herein no later than 30 seconds after application of input voltage and output load.

3.4 Environmental requirements. The rectifier power supply shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.4.1 Vibration. The rectifier power supply shall meet the following vibration requirements:

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3.4.1.1 Vibration, nonoperating. The rectifier power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.4.1.2 Vibration, operating. The rectifier power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The rectifier power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.1.

3.4.2.2 High temperature, operating. The rectifier power supply shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.2.

3.4.2.3 Low temperature, nonoperating. The power supply shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.3.

3.4.2.4 Low temperature, operating. The rectifier power supply shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.4.

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3.5 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. All burrs, sharp edges, and rosin flash shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the rectifier power supplies shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

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4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five rectifier power supplies, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the rectifier power supply by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five rectifier power supplies will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The rectifier power supplies shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each rectifier power supply delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the rectifier power supply shall be subjected to quality conformance tests under the following conditions:

(a) Temperature: 65 degrees F to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

(b) Altitude: Normal ground.

(c) Vibration: None.

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(d) Humidity: Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities, and shall provide personnel for performing the quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 AC power supply - Invertron Model 161A or equivalent.

4.4.3.2 Meters

4.4.3.2.1 Voltmeters

(a) AC high range. 0-150 volts rms at 400 Hz, minimum input impedance 3000 ohms.

(b) AC low range. 0-0.001, 0-0.003, and 0-0.1 volt rms full scale. Frequency response flat from 390 Hz to 150 kHz minimum. Minimum input impedance shall be at least 100 times the terminating or output impedance of the band pass filter specified in 4.4.3.3.

(c) DC 0-150, 0-300, and 0-750 volts dc full scale. Minimum input resistance 500 ohms per volt.

4.4.3.2.2 Ammeters

(a) AC 0-0.5 amperes rms at 400 Hz. Two ac ammeters are required.

(b) DC high range. 0-150 ma, dc.

(c) DC low range. 0-5 ma, dc.

4.4.3.2.3 Ohmmeter. The ohmmeter shall be capable of applying 500 volts dc between two points while the insulation resistance between those two points is being measured. Full scale deflection of the ohmmeter shall be not less than 50 megohms, and not greater than 100 megohms.

4.4.3.3 Band pass filter. The band pass filter shall be used during the measurement of the ac voltage content in

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the rectifier power supply output voltages in a frequency band from 390 Hz to 150 kHz. The purpose of this filter is to transfer maximum voltage, not maximum power, in its pass band. Since the ac output impedance of each rectifier power supply output circuit is relatively low, the inclusion of a simulated generator impedance ahead of the filter shall be allowed if corrections are applied to the test measurements to account for the effect of this impedance. The band pass filter shall have the following characteristics:

4.4.3.3.1 Cut-off frequencies. 375 plus or minus 15 Hz and 160 kHz plus or minus 10 kHz.

4.4.3.3.2 Voltage attenuation in pass band. 0 plus or minus 3 decibels from 390 Hz to 150 kHz.

4.4.3.3.3 Voltage attenuation in rejection regions. 18 decibels per octave reference, with a maximum attenuation of at least 80 decibels required.

4.4.3.3.4 Input impedance. The minimum input impedance of the filter including the simulated generator impedance, if used, shall be not less than 3000 ohms from 390 Hz to 150 kHz.

4.4.3.3.5 DC voltage. The filter shall be capable of withstanding 325 volts dc maximum continuously applied to its input terminals.

4.4.3.3.6 AC voltage. The filter components shall be capable of withstanding the effect produced by placing two volts ac, rms maximum on the filter input terminals.

4.4.3.4 Dummy load resistors

4.4.3.4.1 Section 1, 300 volt dc circuit. Sufficient wire-wound resistors to make up a load of 10,000 ohms plus or minus 2 percent at 10 watts minimum capacity.

4.4.3.4.2 Section 1, 150 volt dc circuit.

(a) 60 ma load. Sufficient resistors to make up a load of 2500 ohms plus or minus 1 percent at 10 watts minimum capacity.

(b) 20 ma load. Sufficient resistors to make up a load of 7500 ohms plus or minus 1 percent at 4 watts minimum capacity.

4.4.3.4.3 Section 2, 150 volt dc circuit. This load shall consist of a fixed resistance and a rheostat connected

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in series. The fixed resistance shall consist of sufficient resistors to make up 1000 ohms plus or minus 10 percent at 20 watts minimum capacity. The total resistance of the rheostat shall be 1000 ohms minimum with a minimum current carrying capacity of 130 ma dc.

4.4.3.4.4 Section 2, 90 volt dc circuit. This load shall consist of a fixed resistance and a rheostat connected in series. The fixed resistance shall consist of sufficient resistors to make up 50,000 ohms plus or minus 10 percent at 1/2 watt minimum capacity. The total resistance of the rheostat shall be 35,000 ohms minimum with a minimum current-carrying capacity of 0.005 ampere dc.

4.4.3.5 Connector, female, MIL-C-24308 (Part Number M24308/1-2)

4.4.3.6 Terminal board. The terminal board shall have 10 double screw-type contacts. There shall be sufficient insulation resistance and creepage distance between adjacent double contacts to withstand 1000 volts rms at 60 Hz for one minute without any arcing or breakdown of insulation. The insulation resistance shall be not less than 1000 megohms at 500 volts dc after this test.

4.4.3.7 Switches

4.4.3.7.1 Three-pole, single-throw. One three-pole, single-throw switch is required. The switch shall have sufficient insulation resistance and creepage distance between adjacent poles to withstand 1000 volts rms at 60 Hz for one minute without any arcing or breakdown of insulation. Each switch pole shall be capable of carrying a one-half ampere minimum resistive load current at 115 volts rms, 400 Hz continuously without any evidence of overheating. Maximum contact drop at one-half ampere rms shall not exceed 0.1 volt rms.

4.4.3.7.2 Single-pole, single-throw. Two single-pole, single-throw switches are required. Each switch shall be capable of carrying 0.2 ampere minimum resistive load current at 150 volts dc continuously without any evidence of overheating. Maximum contact drop at 0.2 ampere dc shall not exceed 0.1 volt dc.

4.4.3.8 High potential test set. The high potential test set shall be capable of furnishing 1000 volts rms at 60 Hz. The test set will be used for insulation testing and shall include an alarm bell or light to indicate when any arcing or insulation breakdown occurs. The test set shall be set to operate at one milliampere for breakdown indication.

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4.4.4 Test setup. See Drawing 738826 for circuit terminal designations. Where connections between the connector specified in 4.4.3.5 and the terminal board specified in 4.4.3.6 are required, keep the leads unbundled and as short as possible. This condition is very important if the output ac voltage content measurements specified in 4.4.5.3.4 are to be accurate.

4.4.4.1 AC input voltage connections. Connect the 115-volt rms, three phase, 400 Hz terminals of the connector to three contacts of the terminal board. Connect any two of these three terminal board contacts to the ac ammeters specified in 4.4.3.2.2(a). Connect the remaining terminal board contact and the remaining terminals of each of the ammeters to the load side of the three-phase switch specified in 4.4.3.7.1. Connect the input side switch terminals to a source of power as specified in 4.4.3.1. The switch shall be in the open position.

4.4.4.2 Section 1, 300 volt dc circuit. Connect the 300-volt dc terminals of the connector to two unused contacts on the terminal board. Connect the dummy load specified in 4.4.3.4.1 between these same contacts on the terminal board.

4.4.4.3 Section 1, 150 volt dc, 60 ma circuit. Connect the positive 150-volt dc, 60 ma terminal of the connector to one unused contact on the terminal board. Connect the dummy load specified in 4.4.3.4.2(a) between this terminal board contact and the same terminal board contact used for the negative terminal of the 300-volt dc circuit.

4.4.4.4 Section 1, 150 volt, 20 ma circuit. Connect the positive 150-volt dc, 20 ma terminal of the connector to one unused contact on the terminal board. Connect the dummy load specified in 4.4.3.4.2(b) between this terminal board contact and the same terminal board contact used for the negative terminal of the 300-volt dc circuit.

4.4.4.5 Section 2, 150 volt dc, 125 ma circuit. Connect the 150-volt dc, 125 ma terminals of the connector to two unused contacts on the terminal board. Connect one switch as specified in 4.4.3.7.2, the dummy load specified in 4.4.3.4.3, and the dc, ma specified in 4.4.3.2.2(b) in series. Connect this string of components between the above specified terminal board contacts. The switch shall be placed nearest the positive output voltage terminal.

4.4.4.6 Section 2, 90 volt dc circuit. Connect the negative 90-volt dc terminal on the connector to the re-

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maining unused contact on the terminal board. Connect one switch as specified in 4.4.3.7.2, the dummy load specified in 4.4.3.4.4, and the dc ma specified in 4.4.3.2.2(c) in series. The negative terminal of milliammeter shall be connected to negative terminal of the 150-volt dc, 125 ma circuit. The switch shall be placed nearest the negative 90-volt dc output terminal.

4.4.5 Test procedure

4.4.5.1 Circuit check. The rectifier power supply shall be wired as specified on Drawing 738826.

4.4.5.2 Insulation tests

4.4.5.2.1 Dielectric test. With all capacitors disconnected from their circuits and after all the rectifiers are shorted by means of shorting cables, apply test voltages to the rectifier power supply terminals as specified in 3.3.3.1. All rectifiers can be shorted by using three independent cables which connect points as follows: Cable No. 1; Points 1, 2, 3, 4, and 7. Cable No. 2; Points 10, 11, and 12. Cable No. 3; Points 13, 14, 15, 16, and 19. No arcing or breakdown of insulation shall occur. Replace all capacitors after this test.

4.4.5.2.2 Insulation resistance. Measure the insulation resistance of the rectifier power supply as specified in 3.3.3.2. The insulation resistance shall be not less than the minimum value specified in 3.3.3.2.

4.4.5.3 Operating tests. Plug the test setup connector specified in 4.4.4 into the rectifier power supply receptacle. Set both dummy load rheostats to their highest resistance settings. Close all switches. Adjust the Section 2, 150-volt dc circuit load rheostat so that the high range milliammeter reads 125 plus or minus 5 ma. Adjust the Section 2, 90-volt dc circuit load rheostat so that the low range milliammeter reads 1.5 plus or minus 0.1 ma.

4.4.5.3.1 Input voltages. Using the voltmeter specified in 4.4.3.2.1(a), measure the three phase, line-to-line, rectifier power supply input voltages at the terminal board. They shall be as specified in 3.3.1.1.

4.4.5.3.2 Input current. With no voltmeters connected to the test setup, read the two input line currents as measured by the test setup ac ammeters. Neither line current shall exceed the limit specified in 3.3.1.3.

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4.4.5.3.3 DC output voltages. Using the dc voltmeter specified in 4.4.3.2.1(c), measure the dc output voltage of each circuit. When the dc output load voltage of the Section 2, 90-volt dc circuit is measured, the voltmeter shall be placed directly across the dummy load and the dummy load rheostat adjusted so that load current is as specified in 3.3.2.2.2.3. All remaining dc output voltage measurements shall be made at the terminal board. The measured voltages shall be as specified in 3.3.2 within 30 seconds after application of input voltage and output load. Note that for the circuits of Section 2, dc output voltage measurements must be made for more than one load condition.

4.4.5.3.4 Total ac voltage content in the output voltages. Open the Section 2, 90-volt dc circuit load switch, and adjust the Section 2, 150-volt dc, 125 ma circuit load to 90 plus or minus 5 ma. Using the band pass filter specified in 4.4.3.3 with the ac voltmeter specified in 4.4.3.2.1(b) connected across the filter's terminating or output impedance, measure at the terminal board the ac voltage from each of the rectifier power supply output circuits. Call these voltages E_M.

Shut off the three-phase rectifier power supply input voltage. Using the same filter-voltmeter setup, measure at the terminal board the ambient electrical noise level for each circuit. Call these voltages E_N.

Where the measured noise (E_N) for a circuit is greater than 20 percent of the ac voltage (E_M) measured for that circuit with power applied to the rectifier power supply, correct the ac voltage readings as follows:

$$\text{Corrected ac voltage} = \sqrt{E_M^2 - E_N^2}$$

If a simulated generator impedance was used for this test, apply the factor necessary to convert the above corrected ac voltage readings to the actual ac voltage content of each circuit tested.

(a) Section 1. The total ac voltage content of each circuit of Section 1, as determined above, shall not exceed the limits specified in 3.3.2.1 for that circuit. Voltage measurements are to be made within 30

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seconds after application of input voltage and output load.

(b) Section 2. The total ac voltage content of each circuit of Section 2. as determined above shall not exceed the limits cited in 3.3.2.2 for that circuit. Voltage measurements are to be made within 30 seconds after application of input voltage and output load.

4.4.5.4 Visual and dimensional examination. Each rectifier power supply shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.5.5 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The rectifier power supply, less pack or packaging, shall be mounted securely to a vibration device by its mounting holes and vibration tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The rectifier power supply shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.1.1.

4.5.1.3 Vibration, operating. The rectifier power supply shall be subjected to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.4.1.2.

4.5.2 Temperature and humidity

4.5.2.1 High temperature and humidity, nonoperating. The rectifier power supply shall be subjected to the "High temper-

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ature and humidity, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.1.

4.5.2.2 High temperature, operating. The rectifier power supply shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.2

4.5.2.3 Low temperature, nonoperating. The rectifier power supply shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.4.2.3.

4.5.2.4 Low temperature, operating. The rectifier power supply shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.5.3 Output impedance. Operate the rectifier power supply under nominal load. Apply not more than 30 volts rms at the frequency or frequencies specified in 3.3.2 through a suitable series-blocking capacitor to the output terminals of the circuit under test. The output impedance of each circuit, as determined by appropriate calculations using the blocking capacitor as reference, shall not exceed the limits specified in 3.3.2.

4.5.4 Section 2, ac voltage content. Operate the rectifier power supply so that full load is drawn from each output circuit of Section 1, 90 plus or minus 5 ma is drawn from the 150-volt dc circuit of Section 2, and no load is drawn from the 90-volt dc circuit of Section 2. Using suitable band pass filters and an amplifier if necessary, measure the ac voltage content in the frequency bands specified in 3.3.2.2 for each circuit. The ac voltage content within the frequency bands listed for each circuit shall not exceed the limits specified in 3.3.2.2.

4.5.5 Energy absorption. Operate the rectifier power supply under full load. Using a suitable dc energy source and a repeat-cycle control circuit, allow energy to be put into the 90-volt dc circuit through its output terminals as specified in 3.3.2.2.2.6. The 90-volt dc circuit output voltage, as determined by metering this output voltage with a calibrated oscilloscope, shall not change by more than the limit specified in 3.3.2.2.2.6.

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4.5.6 Operating life. For this test a new rectifier power supply may be used, or a rectifier power supply that has been under test may be overhauled for the test. Operate the rectifier power supply under full load for a total of 199 plus or minus 1 hour. At the end of each 85 plus or minus 5 minutes of operation, shut off the three-phase input power. Allow the rectifier power supply to return to the ambient temperature before restarting for the next 85 plus or minus 5-minute cycle. At the end of each 10 hours of operation, the rectifier power supply shall meet the tests specified in 4.4.5.3. No maintenance shall be allowed during this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The rectifier power supply shall be packaged in accordance with Method IID of MIL-P-116 and LD 495864.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The rectifier power supply shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The rectifier power supply shall be packed in accordance with LD 495864 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (see 6.2).

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5.2.3 Level B. Not applicable.

5.2.4 Level C. The rectifier power supply shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with the requirements of MIL-STD-129 and LD 495864.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The rectifier power supply is intended for use in Torpedo Mark 37 Mods 2 and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

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6.3 Preparation for delivery. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
Navy - OS
Project No. 1355-N174

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander
Naval Ordnance Systems Command
Maintenance Management and Supporting
Programs Division
Standardization Section
Attn: ORD-05223
Department of the Navy
Washington, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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